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Pillar: Kolmogorov Theory (KT)

A Lean 4 Formalization of Kolmogorov Theory

Axioms, Theorems, and Project Status

Machine-checking the KT corollaries against an explicit AIT interface

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Abstract

We are building `KTAIT`, a Lean 4 + Mathlib formalization whose purpose is narrow and load-bearing: to *machine-check that the Kolmogorov-Theory (KT) corollaries follow logically from an explicit algorithmic-information-theory (AIT) interface*, and that the KT ontology is typed so that structural errors (e.g. comparing a whole pattern to its own part) cannot silently compile. We do *not* re-prove classical AIT: the standard AIT and probability results (the invariance theorem, the coding theorem, symmetry of information, Kraft's inequality, Bayes' rule, the Solomonoff–Levin universal semimeasure, the recursion theorem, Rice's and Chaitin's theorems, and the Vereshchagin–Vitányi structure-function results) are taken as a clearly delimited *axiom layer*. Every *KT* corollary is then *proved* from that layer with no `sorry`, and a toy model witnesses that the axiom layer is jointly satisfiable (so the corollaries are not vacuously true). This working paper states the methodology, fixes the axiom layer, inventories every theorem/lemma across the KT corpus with its current formalization status, and sets a prioritized roadmap.

Keywords: Kolmogorov Theory, algorithmic information theory, Lean 4, Mathlib, formal verification, algorithmic regulator theorem, persistence, self-model.

Classification: MSC 68Q30, 03B35, 68V15 / ACM F.4.1

Contents

1 Introduction and methodology

KT has matured to the point where the dominant risk is no longer in the classical results it invokes, but in whether the *new* KT corollaries are *correctly typed* and *logically entailed* by those results. This project addresses exactly that risk with a proof assistant.[1]

1.1 The one rule

The one rule

Standard AIT (and probability) theorems are allowed to be *axioms*. KT corollaries must be *proved* from those axioms with no `sorry`.

If a KT corollary needs a `sorry`, that is a real gap in the paper. If a classical AIT fact is an axiom, that is honest and expected: we are checking the *dependency structure*, not the AIT foundations.

1.2 Why this is worth doing

The bug class we most want to exclude is *ontological*. For example, writing $\mathrm{I}_K(A : \mathcal{S})$ where $\mathcal{S} \subset A$ is structurally vacuous (a whole versus its own part); the correct object is *temporal*, $\mathrm{I}_K(\mathcal{S}_t : \mathcal{S}_{t+\tau})$ or $K(\mathcal{S}_{t+\tau} \mid \mathcal{S}_t, E, C) \ll K(\mathcal{S}_{t+\tau} \mid C)$. In `KTAIT` the KT roles (*substrate, pattern, self-code, readout, regulator, time*) are *distinct types*; the only bridge to a common carrier is an explicit projection, so the bad version *fails to type-check*. The whole-vs-part error becomes a typed obligation, not a silent mistake.

1.3 Soundness: named hypotheses, not global axioms

A naïve encoding `axiom F : forall (F : AITFrame), P F` asserting a law for *every* frame is *inconsistent* here: one can hand-build a frame violating it (e.g. `slack=0` with mismatched K) and derive `False`, which would make all corollaries vacuously true. We therefore state each AIT fact as a *named hypothesis* (a `Prop` about a frame) and let each corollary consume it. Consequently `#print axioms` on every KT corollary shows only Lean's core axioms (`propext`, `Classical.choice`, `Quot.sound`) — strictly stronger than "rests on named AIT axioms." The toy model (§3) then exhibits a concrete frame satisfying the hypotheses, witnessing joint satisfiability.

1.4 Scope

The current effort is **Level 1** (a typed, axiomatic KT/AIT interface) plus **Level 1.5** (the faithful *probabilistic* ART, including its Bayesian posterior and the coding-theorem bridge). Out of scope for now: a computability-backed K from a concrete universal machine (Level 2), and the differential-geometric (Lie-group / Noether) layer of the world-models paper,[4] which is a separate track.

1.5 Citing the formalization

Papers in the KT corpus cite this development with two macros. `\lean{...}` marks incidental Lean text: a module filename, a definition, one of Lean's core axioms. `\ktait{decl}` is a machine-checked-claim citation: the prose at that point asserts a result and names the KTAIT declaration that proves it, so which claims are machine-checked is a lookup rather than an interpretation — for a reader, and for any tool indexing the corpus. Names inside `\ktait` must be declarations; the guard rejects filenames, paths, and core axioms there. Both macros are enforced: papers registered in `docs/citing-papers.txt` have every cited name resolved against the current declarations by `scripts/check_sync.sh`, which the pre-push hook and CI run, so renaming or restating a theorem cannot silently break a citing paper's claim. The canonical preamble definition is `\newcommand{\ktait}[1]{\texttt{\small #1}}`.

2 The standard AIT / probability axiom layer

These are the results we *assume*. They are classical and not in doubt; collecting them here fixes precisely what the KT proofs are allowed to lean on. (Per K's directive, starting from these as axioms — and similarly for Bayes — is acceptable and intended.)

Axiom layer A — complexity

Invariance theorem. K is machine-independent up to an additive $O(1)$.
Coding theorem (unconditional). $c_1 2^{-K(x)} \le m(x) \le c_2 2^{-K(x)}$, i.e. $-\log_2 m(x) = K(x) \pm O(1)$.
Coding theorem (conditional). $m(x \mid y) \asymp 2^{-K(x|y)}$.
Kraft–McMillan. $\sum_p 2^{-|p|} \le 1$, hence $\sum_x m(x) \le 1$.
Symmetry of information. $\mathrm{I}_K(x{:}y) = K(x)+K(y)-K(x,y) = K(x)-K(x \mid y^*)+O(\log)$; chain rule $K(x,y) = K(x) + K(y \mid x^*) + O(1)$.
Multiplicity compression. $N_{\le L}(x) \ge 2^r \Rightarrow K(x) \le L - r + O(\log L)$.
Data processing (unconditional). $\mathrm{I}_K(z{:}y) \le \mathrm{I}_K(x{:}y) + K(z \mid x^*) + O(\log)$: computing z from x creates no information about y beyond the cost of the transformation. **Self-information.** $K(x) \le \mathrm{I}_K(x{:}x) + O(\log)$.

Axiom layer B — probability / Bayes

Solomonoff–Levin universal semimeasure. $m(x) = \sum_{U(p)=x} 2^{-|p|}$; conditional $m(R \mid W) \asymp 2^{-K(R|W)}$.
Bayes' rule + deterministic likelihood. prior $P(p) = 2^{-|p|}$, likelihood $P(x \mid p) = \mathbf{1}\{U(p) = x\}$, evidence $P(x) = m(x)$.
Excess-length decay. $\Pr\{|p| \ge K(x) + k \mid x\} \le 2C\,2^{-k}$.

Axiom layer C — computability / logic

Kleene's second recursion theorem (quines); **Rice's theorem**; **Chaitin's incompleteness** (T proves no $K(x) > n$ for $n > c_T$; K uncomputable); **Vereshchagin–Vitányi**: the Kolmogorov structure function $h_x(\alpha)$ is uncomputable and minimal sufficient statistics are not locatable.

WP0007 consumes three of these in distinct forms, and `AlgorithmicEmergence` keeps them apart because they are not interchangeable: `KUncomputable` (K is not computable), `KThresholdUndecidable` (the predicate $K(x) < b$ is not *decidable* — it is semidecidable, which is the whole content of the discovery barrier), and `KNotApproximable` (no computable f satisfies $K \le f \le K + c$, which does *not* follow formally from the first as stated). `chaitin_certification_ceiling` consumes the Chaitin bound in the "certified lower bounds are capped" form.

3 Current status of the Lean development

The repository (`https://github.com/giulioruffini/KTAIT`) builds green on Lean 4.31.0 + Mathlib v4.31.0. Modules:

Module	Content	Key results
`Ontology.lean`	six typed KT roles + part-whole guard	`SelfCode` car
`Basic.lean`	`AITFrame`, `IK`, `NMAI`, `condStar`	symmetry as
`Probability.lean`	prefix machine, Bayes posterior, coding thm	Lemma 1 (br
`ART.lean`	`AITProb` (posterior), Theorem 2	probabilistic
`Persistence.lean`	`Pers`, `TemporalSelfInfo`	persistence le
`SelfModel.lean`	`DeltaSelf`, Proposition 3	temporal self-
`SelfModelLimits.lean`	quine floor, diagonalization, Chaitin ceiling	WP0192 Prin
`RegulatorSelection.lean`	`ChainRule`	WP0162 Prop
`CoarseGraining.lean`	`RegSuff`, structure function	WP0193 Thm
`AlgorithmicEmergence.lean`	counting bound, raw-length and threshold barriers	WP0007 Thm
`Contrast.lean`	contrast fibre posterior	WP0162 Q3 (
`ShiftInvariance.lean`	`CheapTransform`, `PairSymmetric`	WP0202 App
`PushPull.lean`	`Plant`, `Admissible`, `InwardOnBoundary`	WP0186 Thm
`ModelOrPay.lean`	`MAICapped`; multiplicity witnesses	WP0203 Prop
`RegulationBalance.lean`	`cIK`, `InfoClosed`, `Localized`, `FlowSplit`	WP0203 v4, t
`GroundedRegulation.lean`	`Grounded`, `residual`, `IKCeiling`, `PairMono`	WP0203 grou
`OrbitLabel.lean`	conserved orbit labels	WP0162 App
`NoetherFlow.lean`	`trajLabel`, flow symmetries	generalized N
`Decoder.lean`	`Achievable`, `Inverts`, `dovetail`	WP0058 Prop
`WriteBack.lean`	`bandwidth`, `DataProcessingCond`	WP0058 Prop
`Retention.lean`	`J`, `qStar`, `FirstCrossing`	WP0058 Prop
`CommonSemantics.lean`	`DataProcessingIK`, `SelfInfo` (in `KTAIT/IS/`)	WP0215 com
`Boundary.lean`	`Machine`, `Reach`, `InternalIso` (in `KTAIT/IS/`)	WP0215 bour
`Unfolding.lean`	`Recurrent`, `Layered` (in `KTAIT/IS/`)	WP0215 finit
`ToyModel.lean`	`Toy`, `ToyMachine`, `ToyGap`	satisfiability v
`PatternPersist.lean`	PP! ontology on `AITFrame`	agenthood pe
`PersistenceFlow.lean`	`flowTuple`, `FlowSplit5`, `CIKCeiling`, `noGapFrame`	APB suite: p
`Localization.lean`	`localization_left/mutual/right`, `CondDataProcessing`	WP0218 loca
`TokenTransport.lean`	`Mode`, `LocalizationProfile`, `LocalizationPlan`, `nwPlan`, `natAbsDiff`	WP0218 canc
`Agentoptosis.lean`	`AgentoptoticEvent`, bearer/target split	WP0207 v0.8
`BadStatements.lean`	guards bite	part-whole; *y*

Proved KT corollaries (no `sorry`; `#print axioms` = Lean core only):

- `PP.token_is_agent`, `PP.class_not_agent_here`, `PP.kind_not_agent_here`, `PP.collective_can_be_agent`, `PP.collective_not_agent_here` — the four cells of the WP0207 square (token a_i, constituent class $\mathcal{T}$, collective M, collective kind $\mathcal{L}(M)$) are all *patterns*; agenthood is decided cell by cell on regulatory evidence and follows from patternhood for none of them. The witnesses exhibit a token that is a genuine agent, recurring kinds that are targets without being agents, and a collective shown both ways — composition settles nothing. Possible because agenthood is a *predicate* on the single carrier `Pattern`, following WP0162's "no agent over and above the pattern"; the per-cell reading is WP0207 v0.8's, after the PP! synchronization review.

- `PP.other_target_not_self_targeting` (general, definitional), `PP.AgentoptoticEvent`, `PP.event_agent_not_sel` `PP.targeted_class_not_agent`, `PP.composition_does_not_imply_agenthood`, `PP.CollectiveSelfTargeting` with its projection lemmas `PP.collective_self_targeting_requires_agent` and `PP.collective_self_targetin` and `PP.terminal_action_underdetermines_target` — the WP0207 v0.8 ontology in `Agentoptosis.lean`: an agentoptotic event requires constituent agenthood and internal selection of the terminal action, and deliberately carries no target or beneficiary field; an other-directed objective is not telehomeostatic for its bearer; naming a pattern as target promotes nothing; higher-scale telehomeostasis is by definition collective agenthood plus a self-targeting collective objective; and the same terminal action arises under objectives with distinct targets, so self-termination alone cannot identify the persistence target.

- `PP.ordinary_retains_exceptional_terminates` — WP0207's nested-mutualism reading as a state-dependence witness: a single class-targeting objective retains the bearer in the ordinary state and terminates it in the exceptional one, so higher-scale targeting is compatible with keeping the constituent alive almost everywhere and agentoptosis is the reversal region. The mutualistic baseline itself is architectural and evolutionary; no generic mutual-benefit theorem is stated.

- `PP.deletion_threshold` — WP0207's boxed decision rule as elementary algebra over $\mathbb{Q}$: for positive costs, $0 < p\,C_{\mathrm{FN}} - (1-p)\,C_{\mathrm{FP}}$ exactly when $p > C_{\mathrm{FP}}/(C_{\mathrm{FP}} + C_{\mathrm{FN}})$. The biological and evolutionary content of WP0207 (Ψ, ΔJ, χ, the filter operator, release, enforceability) is deliberately not formalized.

- `PP.token_not_self_targeting`, `PP.class_is_self_targeting` — telehomeostasis is fixed by the objective alone (WP0162 §3), so a token whose objective proxies its class is a genuine agent that is not telehomeostatic. `PP.agentoptosis_gap` states the conjunction: that gap is the room agentoptosis occupies (WP0207). `PP.token_usually_self_targeting` records the default case: when the objective targets the token itself, it *is* telehomeostatic, so the divergence is the exception and not the rule.

- `PP.collective_not_self_targeting` — the same divergence one grain up: a collective instance whose objective proxies its lineage, with `PP.kind_is_self_targeting` closing the square. This is the snowflake-yeast sign anomaly as a typing fact rather than an exception.

- `PP.persists_iff_nmai` — the module's persistence predicate unfolds definitionally to WP0162's $\mathrm{NMAI}(S_t, S_{t+\tau})$. `PP.Persists` is `Persistence.Persistent` on the pattern's trajectory, so persistence is inherited from `Persistence.lean` rather than restated: rational-valued, normalized, and lag-indexed.

- `PP.agent_has_temporal_self_model` — because `PP.IsAgent` is a positive self-regulation gap Δ_{self} (`SelfModel.DeltaSelf`), Proposition 3 applies to any agent verbatim: its future self-code is cheap given its present organization. Nothing is re-proved; the same named hypotheses are consumed.

- `PP.externally_regulated_not_agent`, `PP.agent_need_not_be_self_targeting` — agenthood is computed from Δ_{self}, never asserted, so a pattern with no gap is refused agenthood; and an agent whose objective targets another pattern is not telehomeostatic.

- `PP.room_not_agent`, `PP.decoupled_not_agent` — locus is *derived* from a directional ablation pair, never asserted, so the externally regulated thermostat room and the decoupled pattern are refused agenthood by computation.

- `AITProb.probabilistic_regulator_theorem` — ART Theorem 2, $\Pr((W, R) \mid x, E) \leq C\, 2^{M(W:R)} 2^{-\Delta}$.

- `AITProb.wrapper_bound` — ART Eq. (6), now *derived* from the coding theorem: $\mathrm{post}(e, x) \leq (1/c_1) 2^{K(x)-K(e)}$.

- `AITProb.probabilistic_regulator_theorem_sharp` — ART Theorem 2 *sharp form* retaining the regulator cost: $\mathrm{post} \leq C\, 2^{M(W:R)} 2^{-\Delta} 2^{-K(R)}$ (no extra hypothesis; the headline drops $2^{-K(R)} \leq 1$ via `probabilistic_regulator_theorem_of_sharp`).

- `AITProb.theorem1_posterior_tilt` — ART Theorem 1, two-sided posterior tilt.

- `AITProb.theorem3_onoff_evidence` — ART Theorem 3, ON/OFF evidence ratio $\asymp 2^{\Delta}$ (log-free form).
- `CoarseGraining.theoremB` / `corollaryB` — WP0193 Theorem B / Corollary B: regulatory coarse-graining is uncomputable, by reduction to Vereshchagin–Vitányi.
- `quine_floor`, `self_prediction_dichotomy`, `chaitin_blocks_minimality` — WP0192 Principle 1 / WP0162 Proposition 4: the three independent obstructions to a complete, exact, certifiable self-model (the middle one is an axiom-free diagonalization).
- `contrast_posterior_ranks_by_complexity` — WP0162 Q3 (Eq. 26): the contrast-fiber posterior ranks by joint simplicity, with no $2^{-\Delta}$ tilt.
- `transform_preserves_IK`, `shift_is_not_a_null`, `cheap_transform_not_null` — WP0202 Appendix A. If $y = T_i(x)$ is computable from x^* plus a short index, then $\mathrm{I}_K(x : T_i x) \geq K(T_i x) - K(\mathrm{code}\, i) - 3\,\mathrm{slack}$: a *cheaply-computable transform preserves algorithmic mutual information.* The corollary names an error that actually shipped into a draft — a **circular shift is not a null** for whole-object MAI, since $K(\sigma_s(x) \mid x^*) \leq K(s) + O(1) = O(\log n)$, so the shifted copy retains nearly all of its algorithmic information with the original. Using it as a surrogate subtracts the signal, and a real screening effect was consequently withdrawn as "indistinguishable from noise". `cheap_transform_not_null` states the contradiction directly: no ε below $K(\sigma_s x) - K(s) - 3\,\mathrm{slack}$ can satisfy `IsNullFor`. (A time-aligned mutual information *rate* is a different object, for which a shift *is* a valid null; that object is not an AIT quantity and is not formalized here. Keeping the two apart is the whole point.) Satisfiability witness: `ShiftFrame`, in which the bound is tight, $\mathrm{I}_K(x : \sigma_s x) = K(x)$.
- `PushPull.theorem1_necessity` — WP0186 Theorem 1. Against a free two-sided disturbance ($\underline{D} < 0 < \overline{D}$), holding a band $[x_-, x_+]$ requires pull authority at the upper edge at least $\overline{D}$ and push authority at the lower edge at least $-\underline{D}$. Its contrapositive, `single_signed_cannot_regulate`, is the paper's headline necessity claim: a single-signed controller cannot hold the band. Theorem 2 (`theorem2_sufficiency`) shows the saturating edge rule attains those conditions, and `regulation_iff` assembles the two halves into the equivalence the paper states in pieces; `inward_iff` is the step that collapses the $\forall D$ condition to its two worst cases. Formalizing them showed these are *not* dynamical theorems: their proofs evaluate the field only at x_- and x_+, so their content is arithmetic on two boundary inequalities. The dynamics enter only through Nagumo's theorem, which is *assumed* (see below) and is never used to derive the sign conditions.
- `PushPull.nonstrict_fails_attraction` and its repair `margin_gives_uniform_decrease` — a defect in WP0186's Theorem 2, and the fix. The first is a *counterexample*: a plant meeting every hypothesis of the sufficiency theorem (band $[-1, 1]$, disturbance range $[-1, 1]$, unit authorities, saturating rule) whose closed-loop field vanishes at $x = 5$ under the admissible realization $D \equiv \overline{D}$, so the constant trajectory $x(t) \equiv 5$ never enters the band. Invariance survives, ultimate boundedness fails: the escape clause "strict inequality whenever $D < \overline{D}$" is vacuous because $D \equiv \overline{D}$ is itself admissible. The repair supplies a strict off-band margin $\varepsilon > 0$, giving $\dot{x} \leq -\varepsilon$ and hence entry in time $\leq (x_0 - x_+)/\varepsilon$. The asymmetry is real: the necessary condition is tight and non-strict, while sufficiency for attraction needs one ε of slack.
- `PushPull.sign_diverse_card_ge_two`, `PushPull.minimal_is_two` — WP0186's cardinality claim, with the distinction Lean forced into the open: the theorems bound *signs*, not counts. A regulating flow set must contain at least one flow of each sign, whence $|S| \geq 2$; and two flows,

one of each sign, achieve it, so the bound is tight. Two pushes do not regulate however many there are.

- `PushPull.restoring_not_single_signed`, `PushPull.restoring_decomposition` — WP0186 Corollary 1. For $\kappa > 0$ the leaky-integrator field $-\kappa(x - x^\star)$ is strictly positive below $x^\star$ and strictly negative above it, so no single sign-constrained flow realizes it; and it decomposes explicitly into a non-negative push active below $x^\star$ and a non-negative pull active above. The single-term restoring law is a push–pull motif in collapsed notation, not a counterexample to one.

- `PushPull.lemma1_gap_identity`, with its existential restatement `lemma1_gap_is_S_existential` — WP0186's complexity-gap lemma, against the opaque `AITFrame` carrier. From (A2)+(A3) (`OffSplit`: $K(x_{\mathrm{OFF}}) = K(D_{\mathrm{pred}}) + K(\eta)$ within slack) and (A4) (`OnResidual`: $K(x_{\mathrm{ON}}) = K(\eta)$ within slack), the gap is $\Delta = S \pm 2\,\mathrm{slack}$ with $S := K(D_{\mathrm{pred}})$: the residual $K(\eta)$, possibly $\Theta(N)$, cancels. The regulator's dynamics play no role — the assumptions, not the mechanism, carry the lemma. `no_structure_no_gap` is the degenerate case: if the predictable component is trivial ($S = 0$) the gap vanishes up to slack, so a regulator that caps the amplitude of unpredictable disturbance claims no model content. **A retraction is recorded with these theorems.** An earlier draft chained the identity into ART to conclude that a sustained Δ *forces* model content $M(W{:}R) \gtrsim \Delta$. That inference is not licensed and the paper no longer makes it; the type-level tell is that I_K does not occur in the statement at all, and no arithmetic on `OffSplit`/`OnResidual` will introduce it. WP0186's Proposition 1 is now a selection claim, which belongs with `RegulatorSelection.lean`.

- `ModelOrPay.simple_regulator_cannot_model` — WP0203 Proposition 1, from the single AIT fact that module assumes, `MAICapped` ($\mathrm{I}_K(x{:}y) \le K(y) + \mathrm{slack}$, a named hypothesis about a frame). If $K(R) + \mathrm{slack} < \Delta$ then $\mathrm{I}_K(W{:}R) < \Delta$ for *every* world: a regulator simpler than the gap it opens cannot carry Δ bits of model. Contrapositively (`model_implies_big`), a pair satisfying the modelling disjunct $\Delta \le \mathrm{I}_K(W{:}R)$ has $\Delta \le K(R) + \mathrm{slack}$ — "model it" implies "be big." The clamp then needs no special pleading: by `clamp_MAI_bounded_by_own_complexity`, $K(R) = O(1)$ alone caps its mutual algorithmic information with every world.

- `ModelOrPay.perpair_does_not_lift` and `multiplicity_cancels_penalty` — WP0203 Proposition 2, the multiplicity objection, as *witnesses* rather than assertions. For every k there is a family of 2^k terms each of weight at most 2^{-k} whose total is exactly 1; and $2^k \cdot 2^{-k} = 1$. A per-pair bound therefore places no useful bound on a sum over a fiber of matching size — the ART per-pair penalty is real and is exactly consumed by the multiplicity. These are counterexamples to the lifting step; they establish no forward tail bound and are not evidence for one.

- `ModelOrPay.invertible_bound_insufficient` — a second *insufficiency* witness. The invertible-coupling inequality $\mathrm{I}_K(W{:}R) \ge K(R) - K(x_{\mathrm{ON}}) - O(\log)$ bounds in $K(R) - K(x_{\mathrm{ON}})$, not in $\Delta = K(x_{\mathrm{OFF}}) - K(x_{\mathrm{ON}})$: the values $K(R) = 1$, $K(x_{\mathrm{ON}}) = 0$, $K(x_{\mathrm{OFF}}) = 100$, $\mathrm{I}_K = 1$ satisfy it while leaving I_K far below the gap. The two bounds coincide exactly when $K(x_{\mathrm{OFF}}) \le K(R)$ (`invertible_reaches_gap_iff`), which is the hypothesis nothing supplies and a clamp violates. What is machine-checked is that the invertible-coupling bound is *insufficient*; nothing here establishes the positive claim it was meant to support.

- `RegulationBalance.balance`, `balance_readout`, `deficit`, `balance_rate_finite`, `gap_forces_exhaust` — WP0203 v2, the regulation balance: the positive replacement for the withdrawn concentration corollary. With Q bundling the regulator's action transcript, its retained private state and the *world exhaust*, and under an information-closure hypothesis $K(y \mid \langle Q, \langle x, R\rangle\rangle) \le \mathrm{slack}$

that makes Q an admissible completion,

$$\Delta \;\le\; \mathrm{I}_K(W{:}R) + \mathrm{cIK}(y{:}Q \mid \langle x, R\rangle) + O(\text{slack}).$$

Equivalently (`deficit`) the episode flow is bounded *below* by $\Delta - \mathrm{I}_K(W{:}R)$: the distinctions removed from the scored readout do not disappear. `gap_forces_exhaust` is the trichotomy — no model content, no world-relevant action transcript and no retained state force the whole gap into the exhaust term, which is the regime the clamp family of `ModelOrPay` occupies. Two scope notes belong with these. The first inequality is an *accounting identity*: it is the chain rule together with a closure condition that defines the exhaust as whatever recovers y, so its content is localization, not depth. And the paper's rate form — a sustained extensive gap needs an extensive flow — is analysis on top of `balance_rate_finite` and is deliberately not formalized. `residual_in_completion` is the supporting step converting "there is a residual" into "the residual sits in Q."

- `RegulationBalance.localized_closure_needs_world_record` and `no_localized_exhaust_of_world_record_in` — WP0203 v4's *localization* clause, which is what stops the balance closing by fiat. If the exhaust were an arbitrary string the choice $\Xi := y$ would satisfy recovery trivially; v4 requires Ξ to be a fixed projection of the realized world-side record S. At frame level that shadow is `Localized` ($K(\Xi \mid S) \le \text{slack}$), and what is machine-checked is the anti-vacuity content: a localized exhaust cannot manufacture recovery the world record does not already support, so if S with the visible episode fails to determine y, no localized Ξ closes the accounting. The physical-state structure — what a world-side record *is*, and that the projection is fixed independently of y — is not expressible in an `AITFrame` and is not formalized.

- `RegulationBalance.cyclic_form` — WP0203 v4's cyclic corollary. A regulator returning its private state to a standard state drops the memory channel, leaving model content, action transcript, and world exhaust: *model it, transmit it, or leave it in the world.*

- `PersistenceFlow.algorithmic_persistence_balance` and its suite — the **Algorithmic Persistence Balance (APB)**, per the KT Persistence Theorem Roadmap v2 (2026-08-12; renames and supersedes the previous day's APAT). The persistence reading of the regulation balance: with completion $(Z^+, \mathcal{W}^+, \sigma, a, \Xi)$ and conditioning on the initial context $\mathsf{Ctx} = (C, Z_t, \mathcal{W}_t)$, information closure and a declared five-way ordered split (`FlowSplit5`) give

$$K(\iota \mid \mathsf{Ctx}) \;\le\; M(\iota{:}Z^+ \mid \cdot) + M(\iota{:}\mathcal{W}^+ \mid \cdot) + M(\iota{:}\sigma \mid \cdot) + M(\iota{:}a \mid \cdot) + M(\iota{:}\Xi \mid \cdot) + O(\text{slack}) :$$

  identity-relevant novelty is predicted-away, kept, consolidated, acted out, or left in the world — never unaccounted. `bounded_persistence_forces_flow` caps the three internal channels by their conditional descriptions (`CIKCeiling`) and forces the remainder through interface flow or world complement; `flow_pigeonhole` is its finite max-form (one flow channel carries half the load; no limsup). `self_code_overload` shows novelty recoverable from the later self-code forces a self-code update of the same size (subadditivity through Z_1), and `bounded_self_code_capacity` caps novelty storable in a budgeted identity at $K_0 + \delta + O(\text{slack})$: a bounded identity cannot be an indefinitely growing memory. Finally `persistence_does_not_imply_gap` is the claim-discipline witness (`noGapFrame`): perfect persistence of an incompressible identity versus annihilation to 0^n has $\mathrm{NMAI}(z, z) = 1$, $\mathrm{NMAI}(z, 0^n) = 0$, yet raw gap $\Delta = K(0^n) - K(z) < 0$ — simple destruction is simpler than survival, so no universal theorem takes a persistence advantage to a positive ART/GART gap. The universal statement is APB's accounting; GART remains the counterfactual refinement when a meaningful positive gap exists.

- `GroundedRegulation.grounded_readout`, `grounded_inequality`, `repaired_corollary`, `repaired_good_regulato` — WP0203's grounded exposition, the repaired static-MAI corollary. A designated reguland trajectory z (ON) and z_0 (OFF) replaces the scored readouts; `Grounded` ties $x \leftrightarrow z$ and $y \leftrightarrow z_0$ up to tolerance ε in algorithmic information distance (`gap_transfer`: the readout gap and the reguland gap $\Delta_Z = Kz_0 - Kz$ then agree up to $2\varepsilon + O(\text{slack})$, which blocks the thermometer hack). With the counterfactual residual $L_Z = K(z_0 \mid \langle z, R\rangle)$ (`residual`),

$$\Delta_Z \;\le\; \mathrm{I}_K(z_0{:}R) + L_Z + O(\text{slack}) \;\le\; \mathrm{I}_K(W{:}R) + L_Z + O(\text{slack}),$$

  so under grounding $\mathrm{I}_K(W{:}R) \ge \Delta - L_Z - 2\varepsilon - O(\text{slack})$: the original Good-Regulator reading is recovered exactly when the readout is grounded and the intervention is counterfactually non-collapsing (L_Z small). The proof spine is `RegulationBalance.balance_readout` stopped at the residual instead of closed by Q.

- `GroundedRegulation.simple_regulator_forces_residual` — the ceiling $\mathrm{I}_K(W{:}R) \le KR +$ slack (`IKCeiling`, standard, named hypothesis) turns the grounded inequality around: $L_Z \ge \Delta_Z - KR - O(\text{slack})$. A fixed simple regulator opening an extensive grounded gap is *forced* into an extensive counterfactual residual — the blind clamp/AC regime classified, and the structural reason no horizon-uniform $\mathrm{I}_K(W{:}R) \asymp \Delta_N$ can hold for fixed finite R: static model content is reusable, cumulative regulation benefit is not.

- `GroundedRegulation.closure_by_construction` — WP0203 v10's Remark (automatic closure). In the deterministic Turing-pair idealization the persistent world description is instantiated in the realized world-side record S and the null readout is computable from it, so $K(y \mid S, C) \le 3\,\text{slack}$ for any visible episode C: an information-closing record exists *by construction*, via the named conditional composition fact `CondCompose` ($K(y|S) \le K(W|S) + K(y|W) + \text{slack}$) and conditioning monotonicity. Closure is therefore automatic in abstract ART and becomes a substantive hypothesis only for physically bounded realizations; localization and minimality retain the substantive bookkeeping role.

- `GroundedRegulation.residual_is_completion` and `grounded_balance` — the bridge back to the balance. Under the reguland-level closure with completion $Q = \langle O_R, \sigma_R, \Xi\rangle$, $|L_Z - \mathrm{cIK}(z_0{:}Q \mid \langle z, R\rangle)| \le 2\,\text{slack}$ (the second direction needs `PairMono`, $K(B \mid C) \le K(\langle A, B\rangle \mid C) + \text{slack}$): the residual is not a new sink but exactly the completion term, so the repaired corollary (when L_Z is small) and the WP0203 balance (where L_Z goes when it is not) are two views of the same accounting. `grounded_balance` is `RegulationBalance.balance` instantiated at $x := z$, $y := z_0$, recorded to show the constructions compose. The asymptotic $(1 - o(1))\Delta$ form and the $\Theta(N)$ readings are analysis on top of these finite inequalities and are deliberately not formalized.

- `GroundedRegulation.exact_counterfactual_balance`, `gart_reguland_from_exact_balance`, `grounded_inequal` `exact_counterfactual_balance_closed` — WP0203 v16's equality-level ledger, the conservation seam. With all quantities conditioned on the declared common frame C and $L_{\mathrm{cf}} = K(\zeta_\varnothing \mid \zeta_R, R, C)$ (`residualC`),

$$\Delta_N^{\zeta} = \mathrm{I}_K(\zeta_\varnothing{:}R \mid C) + L_{\mathrm{cf}} - \mathrm{I}_K(\zeta_R{:}R \mid C) - K(\zeta_R \mid \zeta_\varnothing, R, C) + O(\text{slack}),$$

  a two-sided finite-slack identity assembled from conditional symmetry of information (`CondSI`) and the conditional chain-rule exchange (`CondExchange`) — pure finite-string bookkeeping, deliberately carrying *no* reversibility premise (WP0218 supplies the physical complete-ledger reading in the manuscripts, not an algebraic hypothesis here). The GART inequality is then re-derived as its *projection*: dropping the two regulated-branch correction terms ($K(\zeta_R \mid$

$\zeta_\varnothing, R, C) \geq 0$ and $\mathrm{I}_K(\zeta_R{:}R \mid C) \geq -\text{slack}$, `MutualNonnegC`) and composing with conditional data processing (`DataProcessingC`) gives $\Delta_N^\zeta \leq \mathrm{I}_K(W{:}R \mid C) + L_{\mathrm{cf}} + O(\text{slack})$, so the formal dependency graph matches the manuscript narrative equality → projection → inequality. Substituting `residual_is_completion` at conditioning base $\langle R, C\rangle$ closes the ledger: under reguland-level closure the residual term *is* $\mathrm{cIK}(\zeta_\varnothing{:}Q \mid \zeta_R, R, C)$ — the residual is not a sink but the matched-null information localized in the complementary realized records.

- `GroundedRegulation.self_regulation_forces_self_model` and `self_regulation_forces_self_model_grounde` — WP0203 v13's self-regulation corollary. The query is role-relative: the focal pattern decomposes as $P = R \cup H$ ($R \cap H = \varnothing$) and the reguland projection ζ^P tracks the self-side region H, so the implicit task-relevant self-model content $\mathsf{SM}^{\mathrm{imp}}_{P,N}(R) := \mathrm{I}_K(\zeta^P_\varnothing{:}R \mid C)$ is the `IK` $(z_0{:}R)$ of the grounded inequality read at $z := \zeta^P_R$, $z_0 := \zeta^P_\varnothing$. The reguland form rearranges `grounded_readout` to $\mathsf{SM}^{\mathrm{imp}} \geq \Delta_Z - L_Z - O(\text{slack})$; the grounded form composes with `gap_transfer` to give $\mathsf{SM}^{\mathrm{imp}} \geq \Delta - L_Z - 2\varepsilon - O(\text{slack})$ against the scored gap. The decomposition of P and the word "self" are manuscript semantics layered over the same string inequality — no new ontology, no new axiom, and no claim of explicit symbolic self-representation or self-targeting.

- `Localization.localization_sum_exact`, `localization_delta_exact`, `marginal_delta_exact`, `localization_co` — WP0218's Level-1 algebra. The localization coordinates $L_A = K(A, W) - KW$, I_K, $L_W = K(A, W) - KA$ are an invertible reparameterization of the standard pair complexity profile with $L_A + I_K + L_W = K(A, W)$ exactly; between endpoints the coordinate sum changes exactly by the joint-budget change, equivalently $\Delta K_A + \Delta K_W - \Delta I_K = \Delta K(A, W)$ (the Ebtekar–Hutter two-system decomposition), and a fixed joint budget conserves the sum exactly. Abstract `Int` identities — notation checks, deliberately claiming no new invariant; `localization_mutual_eq_IK` ties the abstract mutual coordinate to the frame's `IK` definitionally.

- `Localization.reversible_complexity_invariance` and `localization_balance_reversible` — the reversible wrapper and WP0218's Algorithmic Localization Balance (reversible specialization). Mutual ε-bit reconstruction plus plain subadditivity both ways gives $|Ky - Kx| \leq \varepsilon + \text{slack}$; applied to a fixed partition's paired endpoints, the localization sum moves by at most $\varepsilon + \text{slack}$ while the marginals redistribute freely.

- `Localization.recoverable_description_overlap`, `recurrent_recoverability_implies_mai_bound`, `recurrent_recoverability_persistence_bound` — WP0218's overlap lemma and its temporal instantiation. From `MutualChain` and the new named standard fact `CondDataProcessing` ($\mathrm{I}_K(D{:}Y) \leq \mathrm{I}_K(X{:}Y) + K(D|X) + \text{slack}$), $\mathrm{I}_K(X{:}Y) \geq KD - K(D|X) - K(D|Y) - 2\,\text{slack}$: a description cheaply recoverable from both sides is paid for by their MAI. Instantiated at pattern descriptions at two times with ε-recoverability of a named D_P, this lower-bounds cross-time MAI and (dividing by the larger endpoint complexity) the WP0216 persistence score. One direction only: no converse extraction of generic MAI is stated, and recurrent recoverability is a sufficient mechanism for persistence, not its definition. The plain-conditioned sibling of `IS.common_semantics_bound` (which conditions on shortest programs); the two consume different named hypotheses and are kept separate.

- `TokenTransport` — WP0218's canonical token transport calculus, finite and Mathlib-free. `cnotAW_involutive`, `cnotWA_involutive`, `swap_involutive`, and `gateFor_involutive` check the reversible gate vocabulary; `canonical_mode_reachable` shows one gate converts any canonical mode $\mathsf{A}(x) = (x, 0)$, $\mathsf{AB}(x) = (x, x)$, $\mathsf{W}(x) = (0, x)$ into any other; `transport_plan_exists` builds the northwest-corner $3{\times}3$ plan between any equal-total profiles; `transport_plan_realizable` realizes any plan entrywise by disjoint token-pair gates (an involution, `applyGates_involutive`, so the circuit is reversible); `diag_le_min`, `exists_min_moved_plan`, and `minimum_moved_mass` give the minimum moved token mass $N - \sum_i \min(p_i, p'_i) = \frac{1}{2}\|p - p'\|_1$, the ℓ_1 identity stated in doubled `Nat` form. Scope guard: canonical independent-token sector only — nothing says an arbitrary pair with a given profile factors into such tokens.

- `IS.common_semantics_bound` — WP0215's common-semantics lower bound. If a finite semantic object F is recoverable from p at cost c_1 and from q at cost c_2, both conditioned on the shortest programs p^*, q^*, then $\mathrm{I}_K(p{:}q) \geq K(F) - c_1 - c_2 - 4\,\mathrm{slack}$. Two descriptions of the same finite semantics cannot be algorithmically independent. The paper's $O(\log n)$ is the explicit `slack` term and the coefficient 4 counts the inequalities the chain actually pays: self-information (`SelfInfo`), data processing twice (`DataProcessingIK`), and one swap (`PairSymmetric`). `IS.common_semantics_bound_raw` is the same bound with the recovery costs left as the frame's own conditional complexities; instantiating F at an intervention-response table $T_D(f)$ gives the observer-relative reading. The paper's D-conditioned form $\mathrm{I}_K(p{:}q \mid D) \geq K(T_D(f) \mid D) - O(\log n)$ is *not* proved here: it needs conditional analogues of the two AIT hypotheses, which are not stated. **Two guards bound the reading.** `IS.common_semantics_does_not_give_structure` exhibits a frame where the bound is attained exactly while $K(p \mid q^*) = 97$ of p's 100 bits remain unrecoverable from q: shared semantics is not shared organization, and near-minimality — a separate hypothesis, not assumed here — would force shared *information*, never shared causal form. `IS.equal_complexity_is_not_shared_information` exhibits two descriptions of equal complexity with $\mathrm{I}_K = 0$, so equal description length is not a weak form of the hypothesis.

- `IS.boundary_eq_of_iso` — WP0215's equivalence hierarchy, the direction that holds: an internal isomorphism restricted to reachable states implies equality of the boundary map, here the full output trace on every finite input word. The converse fails, and not by dead code: `IS.parity4_boundary_eq_parity2` and `IS.parity4_not_internal_iso` give parity computed by a counter modulo four and by a counter modulo two — same boundary, every state of the four-state machine reachable (`IS.parity4_all_reachable`), no isomorphism, because four reachable states do not inject into two. Supporting: `IS.trace_eq_of_iso`, `IS.parity_trace_eq`, `IS.red_out`, `IS.red_step`, `IS.run_append`, `IS.reach_step`, `IS.reach_start`.

- `IS.finite_unfolding` — the correction WP0215 records to an earlier KT reply to the unfolding argument. For every deterministic recurrent machine and every finite horizon T there is an acyclic depth-T layered evaluator computing the same map; the witness is constructed (`IS.unfold`, `IS.unfold_valueAt`). The load-bearing modelling choice is that the state carries everything that changes, weights included, so adaptation is no obstacle: an adapting machine is a machine with a larger state. Acyclicity is typed rather than asserted, each layer producing only the next layer's carrier. `IS.unfolding_layers_eq_state` records the construction's shape definitionally and stops there — this development has no cost model, so no size or depth growth is stated.

- `IS.depth_bounded_ignores_late_input` — the honest negative half. A depth-T layered evaluator is a function of the first T inputs alone, whatever it computes; `IS.evalHorizon_ignores_late_input` is the same for the recurrent machine at a fixed horizon. Hence `IS.no_fixed_unfolding_covers_all_horizons`: for an echo machine no fixed depth and no evaluator of that depth reproduces the map at every horizon, since a candidate of depth T is already wrong at $T+1$. The uniform statement — one acyclic machine, all horizons — is *not written here and cannot be*: the type `Layered X Y T` is indexed by T, so a single term of it has one fixed depth. What is proved is what the paper's argument needs, that the unfolding must be re-chosen as the horizon grows.

- `OrbitLabel.genEnergy_conserved` — WP0162 App. C: generalized energy as a conserved orbit label (axiom-free).
- `boundary_sufficient` — WP0162 App. E: the thin boundary as a compressed sufficient statistic.
- `low_complexity_shrinkage` — ART Theorem A4; `CoarseGraining.theoremA`, `CoarseGraining.existence` — WP0193 Theorem A and Proposition 1.
- `Noether.trajLabel_conserved`, `Noether.conserved_comp_symm` — the abstract core of generalized Noether (Structured Dynamics paper): the trajectory label is conserved by the flow, and symmetries commuting with the flow act on conserved quantities. (Continuous-time analogue of the orbit-label theorem.)
- `Homogeneous.homogeneousSpace` — a transitive symmetry action makes the state space a homogeneous space, $X \simeq G/H_x$ (Theorem A4's algebraic core; the manifold/dimension layer remains future work).
- `Decoder.achievable_computable_of_inverse` / `Decoder.no_universal_decoder` — WP0058 Proposition 1(i)–(ii): a *total* computable write-back encoder that recognizes its own domain would decide membership in the achievable set of the developmental map; hence none exists (`exists_map_without_decoder`), and every realized write-back channel is a partial map whose validity domain must be supplied or learned rather than recognized by the encoder itself. Notable as the one KT corollary here that assumes *no* AIT fact at all — the engine (`achievable_computable_of_inverse`) is unconditional, and only the non-vacuity of the undecidability hypothesis enters as a named `Prop` (`ExistsUndecidableAchievableSet`). Satisfiability witness: `inverts_id`.
- `Decoder.partial_inverse_exists` / `Decoder.achievable_rePred` — the two positive halves of the same proposition, which locate exactly what the impossibility costs. Dovetailing search gives a *partial* inverse that returns a genuine preimage wherever it is defined, so inversion as such is available; what is lost is the ability to *refuse*, since the search diverges off the achievable set. Correspondingly the achievable set of a computable developmental map is recursively enumerable, which machine-checks the informal half of WP0058's proof ("enumerate programs, run D, collect outputs") and leaves only undecidability assumed.
- `WriteBack.channel_bound` — WP0058's channel-bound lemma as a single named result covering both links, $\mathrm{I}_K(a : H' \mid w) \leq \mathrm{I}_K(a : z \mid w) \leq K(z \mid w) + O(\log)$. The right link is `condIK_le_condRight`; the left is data processing, assumed as `DataProcessingCond`. Cite this for the whole chain and `condIK_le_condRight` for the second link alone.
- `WriteBack.darwinian_of_dataProcessing` — a Darwinian regime is exactly the general data-processing hypothesis instantiated at the selection signal, so `Darwinian` assumes nothing beyond `DataProcessingCond` plus the caller's Markov premise that σ is the only route from a into H'.
- `WriteBack.bandwidth_le_cond`, the channel bound, and the regime corollaries — WP0058 Proposition 3. Writing λ_B for the write-back bandwidth (defined below on the write-back program, *not* as $\mathrm{I}_K(a : H' \mid H)$), the master bound $\lambda_B \leq K(H' \mid H) + O(\log)$ holds but is weak in the Darwinian regime, where undirected mutation makes $K(H' \mid H)$ large without transmitting anything acquired. The sharp statement is data processing (`condIK_le_condRight`): since a reaches H' only through a channel variable z, $\lambda_B \leq \mathrm{I}_K(a : z \mid H) \leq K(z \mid H) + O(\log)$. Crucially, the write-back bandwidth is *not* $\mathrm{I}_K(a : H' \mid H)$: that quantity is nonzero even

in a strictly Darwinian lineage, since acquired structure influences *who reproduces*. Mutual information cannot isolate direct write-back — directness is causal, not correlational — so WP0058 conditions on the selection signal, $\lambda_B := \mathrm{I}_K(w : H' \mid H, \sigma)$ on the tagged write-back program $w = C(a)$, which attaches the quantity to the channel rather than inferring it from residual dependence, and blocks $a \to \sigma \to H'$ (`writeBack`). No additive split $\lambda_{\mathrm{sel}} = \mathrm{I}_K(a : H' \mid H) - \lambda_B$ is defined: conditional mutual information is not additively decomposable. Then: in the Darwinian regime there is no write-back program, so $\lambda_B \leq O(\log)$ (`no_program_no_write_back`) — the premise is the *absence of the write-back program*, architectural and posited, since no information measure certifies a missing causal path — while the *total* information crossing is bounded by the selection signal, $\mathrm{I}_K(a : H' \mid H) \leq K(\sigma) + O(\log)$ (`darwinian_bandwidth_le_selection`, zeroth-order): that bound is the substantive half. In the Lamarckian regime $\lambda_B \leq K(w \mid H, \sigma) + O(\log)$ (`lamarckian_bandwidth_le_decoder_image`, first-order) — immediate, since λ_B is *defined* on w. `toyWB_writeback_zero_but_total_positive` machine-checks that the two quantities genuinely differ: a witness with $\lambda_B = 0$ and total $= 3$. Also `decoder_charged` ($K(C) \leq K(H) + O(\log)$, Prop. 2's background-free form) and `trivial_decoder_transmits_nothing` (Corollary 1: a trivial write-back program transmits nothing however much was acquired). Witness `ToyWB` satisfies both named AIT hypotheses *and attains* the Darwinian bound (`toyWB_selection_bound_tight`) — the corollaries are tight, not merely non-vacuous.

- `WriteBack.decoder_charged_rel` — WP0058 Proposition 2 proper, the background-relative form. `decoder_charged` charges the *whole* write-back apparatus to the heritable program, but part of it is supplied by a background B: the cell, the parent, the niche, a teacher, an external training pipeline. Under conditional subadditivity relative to a background (`SubadditivityCondRel`, a named `Prop`) an encoder reconstructible from H and B jointly (`RecoverableFromRel`) costs the lineage only $K(C \mid B) \leq K(H \mid B) + O(\log)$. The hypothesis is strictly weaker than `RecoverableFrom` (`toyWB_background_strictly_weaker`) and the bound is attained away from zero (`toyWB_decoder_charged_rel_tight`); `ToyWB` satisfies it (`toyWB_subadditivityCondRel`). What is *not* shown, here or anywhere in the module: that write-back cannot produce novelty, or that the encoder's representational axes were found by prior selection.

- `WriteBack.retention_le_program` — the retention curve $M(n) = \mathrm{I}_K(w : H_{g+n} \mid H_g)$ (`retention`) is bounded at every lag by $K(w \mid H_g) + O(\log)$: what survives never exceeds what was sent. A corollary of `condIK_le_condRight`.

- `Retention.bangbang_optimal` — the optimal inherited-memory kernel is a *threshold rule*. With payoff $J = \sum_{n<N} q(n)\, v(n)$ linear in a box-constrained kernel $q : \mathbb{N} \to [0, 1]$ (`J`, `Admissible`) and $v(n)$ the net value of still acting on n-generation-old information, every admissible kernel is dominated termwise by $q^*(n) = \mathbb{1}[v(n) > 0]$ (`qStar`, with `qStar_of_pos`, `qStar_of_nonpos`, `qStar_admissible`). If v is antitone the retained set is an initial segment (`optimal_kernel_antitone_is_initial_segment`), so the optimum is "retain for L generations, then drop", and L is the *first crossing*: $q^*(n) = 1 \iff n < L$ (`qStar_eq_one_iff_lt_first_crossing`, `qStar_eq_zero_of_first_crossing_le`) for L satisfying `FirstCrossing`, which exists whenever v is somewhere nonpositive (`exists_firstCrossing`).

- `Retention.persistence_is_first_crossing` — the punchline. Instantiating $v(n) = R(n) - \kappa$ (`netValue`, antitone by `netValue_antitone`) with R the predictive information the environmental feature still carries at lag n and $\kappa > 0$ the per-generation cost of carrying it plus the mismatch loss, the optimum retains exactly the lags where $\kappa < R(n)$. Optimal persistence is therefore the first lag at which R crosses the cost threshold — WP0058's $\tau_{\rho,\theta}$ with $\theta = \kappa$.

The matching relation is the *solution* of a linear retention problem, not a modelling choice, and the threshold is set by the cost.

- `Retention.exponential_crossing` — for $R(n) = R_0 e^{-n/\tau}$ with $R_0, \tau > 0$ and $\kappa > 0$, $\kappa < R(n) \iff n < \tau \log(R_0/\kappa)$, so the optimal kernel is the indicator of that interval (`qStar_exponential`) and the crossing lag is *proportional* to the environmental correlation time at fixed cost ratio (`exponential_persistence_proportional_to_tau`). Non-vacuity: `vTest_antitone`, `vTest_firstCrossing` and `toyRetention_qStar_beats_full` exhibit an antitone net value whose optimum is a nontrivial initial segment and strictly beats full retention ($3 > 2$). What is *not* proved: that any real channel sits at this optimum, that the payoff is linear in q, that costs are constant in n, or that decay is exponential. The empirical matching claim remains Hypothesis 1.
- `persistence_conservation` / `conservation_tradeoff` — WP0162 Proposition 2: the ledger $K(O_W) = \mathrm{I}_K(O_W{:}R) + K(O_W \mid R^*)$ (from symmetry of information), and at fixed $K(O_W)$, maximizing I_K = minimizing the conditional residual.
- `meta_persistence` / `meta_persistence_limit` — WP0162 §6 (meta-persistence): a collective submodel of stable complexity $k > 0$ with a bounded transient $K(S_t^C \mid (S_{t+\tau}^C)^*) \leq L$ (delivered by closure + a Foster–Lyapunov contraction of the constituent flow) has collective persistence $\mathrm{Pers}_C \geq 1 - (L + \mathrm{slack})/k$, hence $\to 1$ as $k \to \infty$ — with *no* shared or compatible objective among the parts (the hypotheses mention only $S^C, k, L, \mathrm{slack}$). Non-vacuity: `toy_meta_persistence_fires`.
- `regulator_selection` — WP0162 Proposition 1: $\arg\min_R K(R \mid W) = \arg\max_R [M(W{:}R) - K(R)]$.
- `regulator_selection_order` — the order form of the same proposition: under the chain rule, minimizing $K(R \mid W)$ is equivalent to maximizing $M(W{:}R) - K(R)$, as an order-equivalence rather than an argmin identity.
- `AlgorithmicEmergence.no_universal_compressor` / `no_universal_compressor_micro` — WP0007's discovery barrier in the form that carries the intuition: no total computable compressor returns, for every string admitting a description shorter than itself, such a description. Asks nothing about optimality. Consumes `CompressibleUndecidable`, the undecidability of $\{x : K(x) < |x|\}$, which is a *different* classical fact from `KThresholdUndecidable` — a diagonal slice of an undecidable set need not be undecidable.
- `AlgorithmicEmergence.no_compression_improver_of_raw` — WP0007's arbitrary-threshold corollary as the paper states it: derived from `no_universal_compressor` by specializing at $b = |x|$, and consuming *no* undecidability input of its own. `no_compression_improver` proves the same conclusion directly from `KThresholdUndecidable`; that route is sound but its hypothesis is not independent.
- `AlgorithmicEmergence.threshold_undecidable_of_compressible` — why. Undecidability of the threshold relation $\{(x, b) : K(x) < b\}$ *follows* from undecidability of its diagonal slice $\{x : K(x) < |x|\}$ along the computable map $x \mapsto (x, |x|)$. The converse fails, which is why the raw-length statement needs its own proof; once the slice is settled the full relation costs nothing further. So the development rests on one classical input here, not two.
- `AlgorithmicEmergence.compressor_of_threshold` — the hierarchy, recorded as a lemma because it is easy to assert backwards. A threshold procedure specializes at $b = |x|$ to a universal compressor, so "no universal compressor" *implies* "no threshold procedure". The raw-length

statement is the stronger one and is **not** a corollary of the arbitrary-b theorem; the converse fails, since a procedure required to work only at $b = |x|$ is a weaker object than one required to work at every b.

- `AlgorithmicEmergence.no_compression_improver` / `no_compression_improver_micro` — WP0007's discovery barrier: no computable procedure returns, for every string (resp. micro-experiment) and threshold b, a description shorter than b whenever one exists. Strictly weaker a demand than optimality, and still impossible: validity already gives $\mathrm{len}(A(x, b)) < b \Rightarrow K(x) < b$, so the threshold guarantee would make the undecidable predicate $K(x) < b$ decidable. The AIT input is `KThresholdUndecidable`. The companion observation — that the predicate is *semidecidable*, so open-ended search finds a shorter description whenever one exists and only fails to halt otherwise — is deliberately not formalized: it would need an operational semantics for programs, a halting-in-t-steps predicate and the dovetailing construction. The theorem rules out a *total* procedure that halts on every input while exploiting every available compression; semidecidability is the complementary positive fact about partial search, so nothing proved here rests on it.

- `AlgorithmicEmergence.identification_barrier` — WP0007's optimality barrier: no computable procedure returns, for every finite micro-experiment, a shortest program for its macrodata. "Reduction is not construction." The reduction runs along the `Faithful` shift-and-read embedding of WP0007 Theorem 1; the AIT input is the uncomputability of K, assumed as the named hypothesis `KUncomputable`. `identification_barrier_conditional` is the same statement with complexity taken conditional on the microlaw and observation map; it is a variant kept for the development's own sake, and is *not* what WP0007's conditional-form corollary says — that is `conditional_complexity_uncomputable` above. WP0007's numbering has shifted three times as results were inserted ahead of others, so entries here name results rather than number them.

- `AlgorithmicEmergence.no_additive_approximation` — WP0007 Proposition 1: relaxing "shortest" to "within a fixed additive constant c of shortest" does not restore computability, for any c. The AIT input here is `KNotApproximable`, which is *not* implied by `KUncomputable` as stated — it is the Berry argument run with slack, proved in WP0007 Proposition 1 and assumed here.

- `AlgorithmicEmergence.counting_bound` / `few_low_complexity_strings` / `most_states_incompressible` — WP0007's residual-information barrier, proved outright rather than assumed: distinct strings need distinct shortest programs, so fewer than 2^m strings have a shortest conditional program of length $< m$, and the complement is therefore more than a $1 - 2^{-d}$ fraction of the state space. The bound holds for every *fixed* conditioning string shared across the ensemble and does not depend on it — no choice of a common microlaw and observation map improves the fraction.

- `AlgorithmicEmergence.conditional_complexity_uncomputable` — WP0007's conditional-form corollary: conditional complexity relative to (D_n, C_n, n) is not computable. The conditioning *varies with the input*, so this does not follow from uncomputability at a fixed conditioning string; the formalization carries the paper's diagonal argument, with the counting witness and the constant-size search as its two named hypotheses.

- `AlgorithmicEmergence.compression_of_good_model` / `sequential_codelength_eq_logloss` — WP0007's bridge from exact description to scientific theory, which is asymmetric. A short program for one record need not be a useful generative model, and that direction is argued in the paper rather than formalized ("useful model" is not a formal predicate). The direction the interpretation rests on is the two-part coding bound: a simple model assigning the data high probability *witnesses* a short description of them, so a predictive macromodel must realize reusable compression and the limits on finding compression bound the substrate every such model needs. The bound is a named hypothesis (classical, Li–Vitányi); the log-loss identity is proved outright, and is the precise sense in which prediction and compression coincide — *sequential probabilistic* compression, not compression in general.

- `AlgorithmicEmergence.chaitin_certification_ceiling` — WP0007 Proposition 2: certified complexity lower bounds are capped at a constant fixed by the theory and the reference machine, so no macromodel longer than that constant is certifiably shortest for any record. Same obstruction as `chaitin_blocks_minimality`, stated for macromodel optimality rather than self-models. Uncomputability is uniform over the family; unprovability is per-instance.
- `algorithmic_emergence` — **superseded.** This declaration stated WP0007's former "no algorithm finds the optimal coarse-graining" result. WP0007 v0.3.0 withdrew that theorem: the objective is degenerate under a constant observation map, the target moves as C varies, and no reduction from the unrestricted structure function into the induced subclass was ever given. The declaration is retained only because `corollaryB` documents the regulatory case, where WP0193 fixes the target and the reduction does go through. Cite `identification_barrier` for WP0007.
- `AITProb.probabilistic_regulator_theorem_conditional` — ART sharp form in the $2^{-K(R|W)}$ reading (posterior favors regulators simple given the world).
- `PrefixMachine.lemma1_posterior_bounds` — the Bayes↔Kolmogorov bridge, $\frac{1}{c_2}2^{K(x)-|p|} \leq P(p \mid x) \leq \frac{1}{c_1}2^{K(x)-|p|}$.
- `self_regulation_temporal_model` — Proposition 3, $K(S_{t+\tau} \mid S_t, E, C) \leq K(S_{t+\tau} \mid C) - \Delta_{\text{self}} + 2\,\text{slack}$.
- `pers_eq_nmai`, `persistent_pos` — persistence as temporal self-information.

Assumed bridges, and what is deliberately not formalized. `PushPull.lean` assumes one classical control-theory fact — *Nagumo's theorem*, that for a locally Lipschitz field on a closed interval an inward-pointing boundary condition is equivalent to forward invariance — as the predicate `NagumoBridge`. This is the dynamical counterpart of not re-proving the coding theorem. It is an assumed bridge, not a KT result, and nothing in the module derives the sign conditions from it. `ModelOrPay.lean` assumes `MAICapped` and nothing else. The headline conjecture it was written to check — that a faithful regulator sustaining Δ_N either has $\mathrm{I}_K(W{:}R) = \Omega(\Delta_N)$ or dissipates $\Omega(\Delta_N)\, k_B T \ln 2$ — is *not stateable* in `AITFrame`: Q_{diss} is a physical quantity and the frame has no bath, no temperature, and no notion of a physical realization of W or R. The same holds for the escape dichotomy, which quantifies over "any physical realization," and for the saturation claim, which quantifies over sinks of finite capacity. We record the absence as a specification of the work remaining — it needs stochastic thermodynamics over a defined coupling — and not as a result. The known theorem in that shape is Shannon-form ($\langle W_{\text{ext}} \rangle \leq -\Delta F + k_B T\, I(W;R)$); whether it admits a single-sequence algorithmic strengthening with I_K in place of I is open.

Satisfiability witnesses: `toy_symmetry` (symmetry of information holds for `Toy`), `toyMachine_coding` (coding theorem holds for `ToyMachine`), `toy_self_model_fires` (the self-model corollary fires with a genuine gap $\Delta_{\text{self}} = 3$). **Guards:** `#check_failure (peers A S)` (part-whole), and `ystar_form_holds` vs `rawy_form_fails` (symmetry must use y^*, not y).

4 Theorem inventory and formalization status

Every numbered/named formal statement across the KT corpus, classified **[STD]** (standard AIT/probability/logic — axiom candidate) or **[KT]** (Kolmogorov-Theory-specific — proof target), with status. Sources: ART paper,[2] Pattern Persistence (WP0162),[3] Agent Know Thyself (WP0192), Regulatory Coarse-Graining (WP0193), world-models paper,[4] and the foundational AIT-of-consciousness paper.[5]

ID (paper)	Statement (abbreviated)	Class	Status
ART paper — The Algorithmic Regulator (Entropy 2026, 28, 257)			
Lemma 1	$P(p \mid x) = 2^{-\lvert p\rvert}/m(x)$; sandwiched by coding thm	STD	**Proved** (`lemma1_...`)
Theorem 2	$\Pr((W,R) \mid x, E) \le C\, 2^{M(W:R)} 2^{-\Delta}$	KT	**Proved** (`prob_reg_thm`)
Theorem 2 (sharp)	retains regulator cost: $\le C\, 2^{M} 2^{-\Delta} 2^{-K(R)}$	KT	**Proved** (`..._sharp`)
Lemma 2	OFF run lower-bounds world: $K(O_{W,\varnothing}) \le K(W) + c_0$	KT	**Axiom** (hyp. `hoff`)
Eq. (6)	wrapper bound post $\le (1/c_1) 2^{K(x)-K(W,R)}$	KT	**Proved** (`wrapper_bound`)
Theorem 1	coupled-pair posterior tilt $\in [\cdot] 2^{K(x)-K(W)-K(R)+M}$	KT	**Proved** (`theorem1_...`)
Theorem 3	$m(O_{W,R})/m(O_{W,\varnothing}) \asymp 2^{\Delta}$ (as-if objective)	KT	**Proved** (`theorem3_...`)
Theorem A4	low-complexity output ⇒ strict tiny shrinkage of $K(W \mid R)$	KT	**Proved** (`low_complexity_shrinkage`)
Defs 1–2	internal model ($M(W{:}R) > 0$); good regulator ($\Delta > 0$)	KT	n/a (encoded)
Thms A1–A3, Lem A1	coding thm (uncond/cond), Kraft counting, excess-length decay	STD	**Axiom**
Pattern Persistence — WP0162			
Proposition 3	self-regulation ⇒ temporal self-model (Eq. 44)	KT	**Proved** (`self_reg_...`)
Proposition 1	regulator selection $\arg\min K(R \mid W) = \arg\max[M - K(R)]$	KT	**Proved** (`regulator_selection`)
Proposition 2	persistence conservation ledger $K(O_W) = \mathrm{I}_K + K(O_W \mid R^*) \pm O(\log)$	KT	**Proved** (`persistence_...`)
Meta-persistence	collective $\mathrm{Pers}_C \ge 1 - (L{+}\mathrm{slack})/k$; parts need not align	KT	**Proved** (`meta_persistence`)
Proposition 4	self-model incompleteness (3 obstructions)	KT	**Proved** (`quine_floor`,...)
PP! ontology	four cells are patterns; agenthood per cell, composition settles nothing	KT	**Proved** (`PP.token_is_agent`, `PP.collective_not_agent_here`,...)
WP0162 §3	telehomeostasis is fixed by the objective alone	KT	**Proved** (`PP.class_is_self_targeting`)
WP0207	agentoptosis gap: agent ∧ not telehomeostatic	KT	**Proved** (`PP.agentoptosis_gap`)
WP0207 v0.8	event requires agenthood + internal selection; no target/beneficiary field	KT	**Proved** (`PP.AgentoptoticEvent`, `PP.other_target_not_self_targeting`)

ID (paper)	Statement (abbreviated)	Class	Status
WP0207 v0.8	target/composition promote nothing; collective telehomeostasis conditional	KT	**Proved** (`PP.targeted_class_not_agent`, `PP.composition_does_not_imply_agen` `PP.CollectiveSelfTargeting`)
WP0207 v0.8	terminal action underdetermines target	KT	**Proved** (`PP.terminal_action_underdetermines`
WP0207 v0.8	one other-directed objective: retains in ordinary state, terminates in exceptional	KT	**Proved** (`PP.ordinary_retains_exceptional_te`
WP0207 v0.8	deletion threshold $0 < p\,C_{\mathrm{FN}} - (1-p)\,C_{\mathrm{FP}} \iff p > \vartheta$	KT	**Proved** (`PP.deletion_threshold`)
WP0162 §5	agenthood is a positive Δ_{self}, computed not asserted	KT	**Proved** (`PP.externally_regulated_not_agent`)
WP0162 §4	persistence unfolds to NMAI, inherited from `Persistence`	KT	**Proved** (`PP.persists_iff_nmai`)
Proposition 3 (PP)	every agent has a temporal self-model	KT	**Proved** (`PP.agent_has_temporal_self_model`)
WP0207	agent yet not telehomeostatic	KT	**Proved** (`PP.agent_need_not_be_self_targetin`
Orbit-label thm	conserved orbit label = "generalized energy"	KT	**Proved** (`genEnergy_conserved`)
Agent Know Thyself — WP0192			
Proposition 1	temporal self-model (= WP0162 Prop. 3)	KT	**Proved** (same)
Principle 1	self-model incompleteness (= WP0162 Prop. 4)	KT	**Proved** (Phase 4)
Conjecture	acquired self-model prior	KT	n/a (conjecture)
Regulatory Coarse-Graining — WP0193			
Theorem B	targeted sufficient projection uncomputable	STD	**Proved** (`theoremB`)
Corollary B	regulatory coarse-graining uncomputable	KT	**Proved** (`corollaryB`)
Lemma 1	uncomputability of $h_{x \to y}$	STD	**Axiom** (via V–V)
Proposition 1	existence of a bounded regulation-sufficient projection	KT	**Proved** (`existence`)
Theorem A	selection uncomputability	KT	**Proved** (`theoremA`)
Cor. C1–C3	blanket uncomputability/fuzziness; acquisition; discovery	KT	n/a (interp.)
Inheritance Architecture — WP0058			
Prop. 1(i)	no total, domain-recognizing write-back encoder	KT	**Proved** (`no_universal_decoder`)
— engine	a total such inverse would decide achievability	KT	**Proved** (`achievable_computable_of_inverse`)
Prop. 1(ii)	some computable D admits no such encoder	KT	**Proved** (`exists_map_without_decoder`)
— non-vacuity	some computable D has an undecidable achievable set	STD	**Axiom** (`ExistsUndecidableAchievableSet`)
Prop. 1(iii)	a *partial* recursive inverse exists, with domain exactly $\mathcal{A}$	KT	**Proved** (`partial_inverse_exists`, `dovetail`)
— $\mathcal{A}$ is r.e.	the achievable set of a computable D is recursively enumerable	KT	**Proved** (`achievable_rePred`)

ID (paper)	Statement (abbreviated)	Class	Status
— unbounded in time	the dovetailing search has no bound on its running time	KT	n/a (resource claim; no cost model in this development)
Prop. 2	heritable burden net of background, $K(C \mid B) \leq K(H \mid B) + O(\log)$	KT	**Proved** (`decoder_charged_rel`)
— background-free form	$K(C_{\mathcal{D}}) \leq K(H) + O(\log)$	KT	**Proved** (`decoder_charged`)
— cond. sub-add. rel. B	$K(x \mid B) \leq K(x \mid \langle y, B\rangle) + K(y \mid B) + O(\log)$	STD	**Axiom** (`SubadditivityCondRel`)
Channel bound (lemma)	both links: $\mathrm{I}_K(a : H' \mid w) \leq \mathrm{I}_K(a : z \mid w) \leq K(z \mid w) + O(\log)$	KT	**Proved** (`channel_bound`)
— left link	data processing under the Markov premise $a \to z \to H'$ given w	STD	**Axiom** (`DataProcessingCond`)
— right link	$\mathrm{I}_K(a : z \mid w) \leq K(z \mid w) + O(\log)$; info via the channel only	KT	**Proved** (`condIK_le_condRight`)
Master bound	$\lambda_B \leq K(H' \mid H) + O(\log)$; the acquired state drops out	KT	**Proved** (`bandwidth_le_cond`)
Prop. 3 (Darwinian, λ_B)	write-back channel empty: $\lambda_B = \mathrm{I}_K(a : H' \mid H, \sigma) \leq O(\log)$	KT	**Proved** (`no_program_no_write_back`)
Prop. 3 (Darwinian, λ_{sel})	zeroth-order: total $\mathrm{I}_K(a : H' \mid H) \leq K(\sigma)$	KT	**Proved** (`darwinian_bandwidth_le_selection`)
— regime from data proc.	`Darwinian` is `DataProcessingCond` at the selection signal	KT	**Proved** (`darwinian_of_dataProcessing`)
Prop. 3 (Lamarckian)	first-order: $\lambda_B \leq K(C(a) \mid H)$, the decoder image	KT	**Proved** (`lamarckian_bandwidth_le_decoder_im`
— joint $\geq$ marginal	$K(x \mid z) \leq K(\langle x, y\rangle \mid z) + O(\log)$	STD	**Axiom** (`JointGeMarginal`)
— cond. sub-additivity	$K(x) \leq K(x \mid y) + K(y) + O(\log)$	STD	**Axiom** (`SubadditivityCond`)
— conditioning $\leq$ uncond.	$K(x \mid y) \leq K(x) + O(\log)$	STD	**Axiom** (`CondLeUncond`)
Corollary 1	write-back is capped by the write-back program the encoder can form	KT	**Proved** (`trivial_decoder_transmits_nothing`)
Retention bound	$M(n) = \mathrm{I}_K(w : H_{g+n} \mid H_g) \leq K(w \mid H_g) + O(\log)$	KT	**Proved** (`retention_le_program`)
Prop. 4(i)	linear box-constrained kernel: the optimum is bang-bang	KT	**Proved** (`bangbang_optimal`)
Prop. 4(ii)	antitone net value $\Rightarrow$ retained set is $[0, L)$	KT	**Proved** (`optimal_kernel_antitone_is_initial`
— L is first crossing	$q^*(n) = 1 \iff n < L$ for `FirstCrossing` L	KT	**Proved** (`qStar_eq_one_iff_lt_first_crossing` `qStar_eq_zero_of_first_crossing_le`
— existence of L	a first crossing exists if v is somewhere non-positive	KT	**Proved** (`exists_firstCrossing`)
— threshold rule	q^* well defined and admissible	KT	**Proved** (`qStar_of_pos`, `qStar_of_nonpos`, `qStar_admissible`)

ID (paper)	Statement (abbreviated)	Class	Status
Prop. 4(iii)	$v = R - \kappa$: optimum retains exactly where $\kappa < R(n)$, i.e. $\tau_{\rho,\theta}$	KT	**Proved** (`persistence_is_first_crossing`, `netValue_antitone`)
Prop. 4(iv)	$\kappa < R_0 e^{-n/\tau} \iff n < \tau \log(R_0/\kappa)$	KT	**Proved** (`exponential_crossing`, `qStar_exponential`)
— proportionality	crossing lag $\propto \tau$ at fixed cost ratio	KT	**Proved** (`exponential_persistence_proportion`)
— non-vacuity	an antitone v whose optimum is a nontrivial initial segment	KT	**Proved** (`vTest_antitone`, `vTest_firstCrossing`)
Hypothesis 1	persistence matching, $\lambda_P^* \sim \tau_E$, in real lineages	KT	n/a (conjecture; the *optimization* is proved above)
Push–Pull Motifs — WP0186			
Theorem 1	opposed flows necessary: $H(x_+) \geq \overline{D}$ and $G(x_-) \geq -\underline{D}$	KT	**Proved** (`theorem1_necessity`)
— contrapositive	a single-signed controller cannot hold the band	KT	**Proved** (`single_signed_...`)
— boundary step	the $\forall D$ condition collapses to two worst cases	KT	**Proved** (`inward_iff`)
Nagumo's theorem	inward on $\partial B \Leftrightarrow$ forward invariance	STD	**Axiom** (`NagumoBridge`)
Theorem 2	saturating edge rule renders the band invariant	KT	**Proved** (`theorem2_sufficiency`)
Thms 1+2	sharp form: invariance $\Leftrightarrow$ both edge conditions	KT	**Proved** (`regulation_iff`)
Remark 2	non-strict off-band bound does *not* give attraction (counterexample)	KT	**Proved** (`nonstrict_fails_...`)
— repair	strict margin ε: $\dot{x} \leq -\varepsilon$, finite-time entry	KT	**Proved** (`margin_gives_...`)
Minimality	sign-diverse $\Rightarrow |S| \geq 2$; two flows attain it	KT	**Proved** (`sign_diverse_...`, `minimal_is_two`)
Corollary 1	$-\kappa(x - x^\star)$ is not single-signed; splits into push + pull	KT	**Proved** (`restoring_not_...`, `restoring_decomp...`)
App. Lemma 1	gap identity $\Delta = S \pm O(\log N)$ from (A2)–(A4)	KT	**Proved** (`lemma1_gap_...`)
— (A2)–(A4)	OFF splits as $K(D_{\text{pred}}) + K(\eta)$; ON tracks η	STD	**Axiom** (`OffSplit`, `OnResidual`)
Remark 7	$S = 0 \Rightarrow$ no gap (pure noise buys no model content)	KT	**Proved** (`no_structure_no_gap`)
Prop. 1 (withdrawn)	sustained Δ *forces* $M(W{:}R) \gtrsim \Delta$	KT	n/a (retracted; not entailed)
The ART Tail Does Not Follow — WP0203			
Proposition 1	$\mathrm{I}_K(W{:}R) \leq K(R) + O(\log)$: a cheap regulator cannot model	KT	**Proved** (`simple_regulator_...`, `model_implies_big`)
— clamp	$K(R) = O(1) \Rightarrow \mathrm{I}_K(W{:}R) = O(1)$ for *every* world	KT	**Proved** (`clamp_MAI_bounded_...`)

ID (paper)	Statement (abbreviated)	Class	Status
— MAI cap	$\mathrm{I}_K(x{:}y) \le K(y) + O(\log)$	STD	**Axiom** (`MAICapped`)
Proposition 2	per-pair 2^{-k} does not lift to a fiber of size 2^k (witness)	KT	**Proved** (`perpair_does_not_lift`, `multiplicity_...`)
Invertible coupling	$\mathrm{I}_K \ge K(R) - K(x_{\mathrm{ON}})$ holds with $\mathrm{I}_K \ll \Delta$ (witness of insufficiency)	KT	**Proved** (`invertible_bound_...`)
— when it reaches Δ	exactly when $K(x_{\mathrm{OFF}}) \le K(R)$	KT	**Proved** (`invertible_reaches_...`)
"Model it or pay for it"	$\mathrm{I}_K = \Omega(\Delta_N)$ or $Q_{\mathrm{diss}} = \Omega(\Delta_N) k_B T \ln 2$	KT	n/a (conjecture; not stateable in `AITFrame`)
Gap transfer	grounding $\Rightarrow \lvert\Delta - \Delta_Z\rvert \le 2\varepsilon + O(\mathrm{slack})$	KT	**Proved** (`gap_transfer`)
Grounded inequality	$\Delta_Z \le \mathrm{I}_K(z_0{:}R) + L_Z \le \mathrm{I}_K(W{:}R) + L_Z + O(\mathrm{slack})$	KT	**Proved** (`grounded_readout`, `grounded_inequality`)
Repaired corollary	grounded, non-collapsing: $\mathrm{I}_K(W{:}R) \ge \Delta - L_Z - 2\varepsilon - O(\mathrm{slack})$	KT	**Proved** (`repaired_corollary`, `repaired_good_regulator`)
Forced residual	$\mathrm{I}_K \le KR + \mathrm{slack} \Rightarrow L_Z \ge \Delta_Z - KR - O(\mathrm{slack})$	KT	**Proved** (`simple_regulator_forces_residual`)
Residual = completion	closure: $\lvert L_Z - \mathrm{cIK}(z_0{:}Q \mid \langle z, R\rangle)\rvert \le 2\,\mathrm{slack}$	KT	**Proved** (`residual_is_completion`, `grounded_balance`)
Exact balance (v16)	$\Delta_N^{\zeta} = \mathrm{I}_K(\zeta_{\varnothing}{:}R \mid C) + L_{\mathrm{cf}} - \mathrm{I}_K(\zeta_R{:}R \mid C) - K(\zeta_R \mid \zeta_{\varnothing}, R, C) \pm O(\mathrm{slack})$	KT	**Proved** (`exact_counterfactual_balance`)
— GART as projection	drop regulated-branch terms + data processing $\Rightarrow \Delta_N^{\zeta} \le \mathrm{I}_K(W{:}R \mid C) + L_{\mathrm{cf}} + O(\mathrm{slack})$	KT	**Proved** (`gart_reguland_from_exact_balance`, `grounded_inequality_from_exact_bal`
— closed ledger	closure: residual term $= \mathrm{cIK}(\zeta_{\varnothing}{:}Q \mid \zeta_R, R, C)$ in the equality	KT	**Proved** (`exact_counterfactual_balance_close`
Implicit self-model	self-regulation query: $\mathsf{SM}^{\mathrm{imp}} \ge \Delta_Z - L_Z - O(\mathrm{slack})$; grounded: $\ge \Delta - L_Z - 2\varepsilon - O(\mathrm{slack})$	KT	**Proved** (`self_regulation_forces_self_model`, `..._grounded`)
Automatic closure	abstract ART: world record closes the accounting by construction	KT	**Proved** (`closure_by_construction`)
— asymptotic form	$\mathrm{I}_K(W{:}R) \ge (1 - o(1))\Delta$ for $\varepsilon, \lambda = o(\Delta)$	KT	n/a (analysis on top of the finite inequalities)
Algorithmic Persistence Balance — roadmap v2 (PersistenceFlow)			
APB	$K(\iota \vert \mathsf{Ctx}) \le \sum_5 M(\iota{:}\cdot \mid \cdots) + O(\mathrm{slack})$	KT	**Proved** (`algorithmic_persistence_balance`)
Forces flow	bounded $Z^+, \mathcal{W}^+, \sigma$ channels force $J_a + J_{\Xi}$	KT	**Proved** (`bounded_persistence_forces_flow`)
Flow pigeonhole	$2\max\{J_a, J_{\Xi}\} \ge \nu - \sum d - O(\mathrm{slack})$, finite	KT	**Proved** (`flow_pigeonhole`)
Self-code overload	recoverable novelty forces $K(Z_1 \vert Z_0, C) \ge \nu - \delta - O(\mathrm{slack})$	KT	**Proved** (`self_code_overload`)
Self-code capacity	budget K_0 caps storable novelty at $K_0 + \delta + O(\mathrm{slack})$	KT	**Proved** (`bounded_self_code_capacity`)
No positive gap	witness: $\mathrm{NMAI} = 1$ persistence, $\Delta < 0$ raw gap	KT	**Proved** (`persistence_does_not_imply_gap`)

ID (paper)	Statement (abbreviated)	Class	Status
— rate corollary	asymptotic $\nu_P \le r_Z + r_M + r_\sigma + r_a + r_\Xi$	KT	n/a (analysis on top of the finite forms)
— identity-grounded bridge	restricted persistence-to-GART score	KT	n/a (readout definition open; roadmap §3 Cor. 2 is existing GART re-instantiated)
Algorithmic localization dynamics — WP0218 (Localization, TokenTransport)			
Sum identity	$L_A + I_K + L_W = K(A, W)$, exact on abstract coordinates	KT	**Proved** (`localization_sum_exact`)
Delta identities	coordinate-sum change $= \Delta K(A, W)$; $\Delta K_A + \Delta K_W - \Delta I_K = \Delta K(A, W)$	KT	**Proved** (`localization_delta_exact`, `marginal_delta_exact`)
Exact conservation	fixed joint budget $\Rightarrow$ conserved coordinate sum	KT	**Proved** (`localization_conserved_exact`)
Reversible invariance	mutual ε-reconstruction: $\lvert Ky - Kx\rvert \le \varepsilon +$ slack	STD	**Proved** (`reversible_complexity_invariance`)
ALB (reversible)	localization sum moves $\le \varepsilon +$ slack under fixed partition	KT	**Proved** (`localization_balance_reversible`)
— cond. data processing	$\mathrm{I}_K(D{:}Y) \le \mathrm{I}_K(X{:}Y) + K(D\|X) +$ slack	STD	**Axiom** (`CondDataProcessing`)
Overlap lemma	$\mathrm{I}_K(X{:}Y) \ge KD - K(D\|X) - K(D\|Y) - 2\,$slack	KT	**Proved** (`recoverable_description_overlap`)
Recurrent recoverability	ε-recoverable D_P at two times bounds cross-time I_K, Pers	KT	**Proved** (`recurrent_recoverability_implies_m` `..._persistence_bound`)
Gates involutive	$C_{A\to W}$, $C_{W\to A}$, SWAP, and mode gates self-inverse	KT	**Proved** (`cnotAW_involutive`, `cnotWA_involutive`, `swap_involutive`, `gateFor_involutive`)
Mode reachability	one gate converts any canonical mode into any other	KT	**Proved** (`canonical_mode_reachable`)
Transport plans exist	equal-total profiles admit a 3×3 plan (north-west corner)	KT	**Proved** (`transport_plan_exists`)
Plans realizable	disjoint token-pair gates realize every entry, reversibly	KT	**Proved** (`transport_plan_realizable`, `applyGates_involutive`)
Minimum moved mass	$N - \sum_i \min(p_i, p'_i) = \frac{1}{2}\lVert p - p'\rVert_1$ (doubled `Nat` form)	KT	**Proved** (`diag_le_min`, `exists_min_moved_plan`, `minimum_moved_mass`)
— canonical realizability	incompressible-block representative of any profile	KT	n/a (standard AIT existence argument; manuscript Prop.)
— token–profile bridge	$O(\log N)$ translation from token lengths to K coordinates	KT	n/a (manuscript-level; deliberately not formalized)

ID (paper)	Statement (abbreviated)	Class	Status
The Simulation Is Phenomenally Itself — WP0215			
Common semantics	$\mathrm{I}_K(p{:}q) \geq K(F) - c_1 - c_2 - 4\,\text{slack}$ when F is cheaply recoverable from both	KT	**Proved** (`common_semantics_bound`)
— cost-free form	same, with the frame's own $K(F \mid p^*), K(F \mid q^*)$	KT	**Proved** (`common_semantics_bound_raw`)
— conditioned on D	$\mathrm{I}_K(p{:}q \mid D) \geq K(T_D(f) \mid D) - O(\log n)$	KT	**TODO** (needs conditional forms of both hypotheses)
— data processing	$\mathrm{I}_K(z{:}y) \leq \mathrm{I}_K(x{:}y) + K(z \mid x^*) + O(\log)$	STD	**Axiom** (`DataProcessingIK`)
— self-information	$K(x) \leq \mathrm{I}_K(x{:}x) + O(\log)$	STD	**Axiom** (`SelfInfo`)
— not structure	bound attained while $K(p \mid q^*) = 97$: shared semantics, no shared form	KT	**Proved** (`common_semantics_does_not_give_str`
— equal size is not it	equal K, zero I_K, in a frame satisfying all three laws	KT	**Proved** (`equal_complexity_is_not_shared_inf`
Equivalence hierarchy	internal isomorphism on reachable states $\Rightarrow$ equal boundary	KT	**Proved** (`boundary_eq_of_iso`)
— converse false	mod-4 and mod-2 parity: same trace, no isomorphism	KT	**Proved** (`parity4_boundary_eq_parity2`, `parity4_not_internal_iso`)
— no dead code	every state of the four-state machine is reachable	KT	**Proved** (`parity4_all_reachable`)
Finite unfolding	any recurrent machine, any horizon T: an acyclic depth-T evaluator agrees	KT	**Proved** (`finite_unfolding`)
— shape of it	T layers, each a copy of the state; no asymptotic claim	KT	**Proved** (`unfolding_layers_eq_state`)
— horizon bound	a depth-T evaluator reads only the first T inputs	KT	**Proved** (`depth_bounded_ignores_late_input`, `evalHorizon_ignores_late_input`)
— no fixed cover	one evaluator of depth T already fails at horizon $T+1$	KT	**Proved** (`no_fixed_unfolding_covers_all_hori`
— uniform version	one acyclic machine, all horizons	KT	n/a (unwritable: `Layered X Y T` is indexed by T)
— unfolding size	growth of the unfolded family with T	KT	n/a (resource claim; no cost model in this development)
Realization relation	$P \models_{\rho,C} \mathfrak{S}$ with commuting and counterfactual conditions	KT	**TODO** (not yet formalized)
Structured experience	$\mathcal{S}(P) \cong \mathfrak{S}(P)$, experience as implemented structure	KT	n/a (not a proof target; no formal referent for experience)
World models / Lie groups — Entropy 2025 (separate geometry track)			
Thms A1–A5	Lie transitivity, homogeneous space, moduli stack	STD	n/a (out of scope v1)
Noether (gen.)	continuous symmetry $\Rightarrow$ conserved quantity (abstract flow core)	KT	**Proved** (`trajLabel_conserved`)

ID (paper)	Statement (abbreviated)	Class	Status
Homogeneous space (transitive action ⇒ G/H)	—	STD	**Proved** (`homogeneousSpace`)
Lie pseudogroups / moduli stacks / Mostow / dim formula	—	geom	**TODO** (not in Mathlib)
Foundational — nix019 / K-input			
Hyp. 1–2, Cons. 1–2	simplicity selection; structured experience; high MAI	KT	n/a (informal)

Legend. Proved: machine-checked in Lean, no `sorry`. **Axiom**: assumed (standard result, or a named hypothesis consumed by a corollary). **TODO**: a KT proof target not yet formalized. n/a: definition, conjecture, interpretive remark, or informal claim (not a proof target in the current scope).

5 Roadmap

Priorities are ordered to maximize KT-correctness assurance per unit effort, exploiting the fact that the AIT/Bayes layer (§2) may be assumed.

Phase 1 — close the ART chain. DONE.
The wrapper bound Eq. (6) is now a *derived* theorem (`wrapper_bound`) from the coding theorem (`CodingLB`); ART **Theorem 1** (`theorem1_posterior_tilt`) and **Theorem 3** (`theorem3_onoff_evidence`, log-free form) are proved; `probabilistic_regulator_theorem` now consumes `wrapper_bound` and rests on the coding theorem + Lemma 2. All four `#print axioms` = Lean core only.

Phase 2 — persistence conservation (WP0162 Prop. 2). DONE.
Ledger $K(O_W) = \mathrm{I}_K(O_W{:}R) + K(O_W \mid R^*)$ (`persistence_conservation`, from symmetry of information) and the trade-off "max I_K = min conditional residual" (`conservation_tradeoff`).

Phase 3 — regulator selection (WP0162 Prop. 1). DONE.
$\arg\min_R K(R \mid W) = \arg\max_R [M(W{:}R) - K(R)]$ proved (`regulator_selection`) from the chain rule; the ART sharp form's $2^{-K(R|W)}$ reading is `probabilistic_regulator_theorem_conditional`.

Phase 4 — self-model incompleteness. DONE.
The three independent obstructions: the quine floor (`quine_floor`, $|Q_A| \geq K(A)$), the self-prediction dichotomy (`self_prediction_dichotomy`, an axiom-free diagonalization), and the Chaitin ceiling on minimality (`chaitin_blocks_minimality`).

Phase 5 — coarse-graining uncomputability (WP0193). DONE.
Theorem B and Corollary B (regulatory coarse-graining uncomputable) proved by an abstract reduction to Vereshchagin–Vitányi (`theoremB`, `corollaryB`).

All five roadmap phases are complete. The WP0162 / WP0192 / WP0193 KT corollaries and the full probabilistic ART layer are machine-checked, `sorry`-free, resting only on Lean core plus the named AIT axiom layer of §2.

Later — geometry track.
The Lie-group/Noether layer of the world-models paper is a distinct development (differential geometry in Mathlib), deferred until the AIT/KT core is complete.

Each phase keeps the invariant: KT results `sorry`-free; `#print axioms` = Lean core + named hypotheses; new hypotheses discharged by an extended toy model.

6 Conclusion

We have a green, machine-checked Level-1/1.5 core: the probabilistic ART (Theorem 2), the Bayes↔Kolmogorov bridge (Lemma 1), the temporal self-model (Proposition 3), and persistence, all proved from an explicit, satisfiable AIT/Bayes axiom layer, with the KT ontology typed so the whole-vs-part error cannot compile. This document is the living status tracker; the roadmap above closes the remaining KT corollaries paper-by-paper while keeping the soundness invariants intact.

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